photinos, for the gluons there should be gluinos, for the W and Z bosons there should be winos and zinos.

On closer inspection, then, supersymmetry seems to be a terribly uneconomical feature; it requires a whole slew of additional particles that wind up doubling the list of fundamental ingredients. Since none of the superpartner particles has ever been detected, you would be justified to take Rabi's remark from Chapter 1 regarding the discovery of the muon one step further, declare that "nobody ordered supersymmetry," and summarily reject this symmetry principle. For three reasons, however, many physicists believe strongly that such an out-of-hand dismissal of supersymmetry would be quite premature. Let's discuss these reasons.

The Case for Supersymmetry: Prior to String Theory

First, from an aesthetic standpoint, physicists find it hard to believe that nature would respect almost, but not quite all of the symmetries that are mathematically possible. Of course, it is possible that an incomplete utilization of symmetry is what actually occurs, but it would be such a shame. It would be as if Bach, after developing numerous intertwining voices to fill out an ingenious pattern of musical symmetry, left out the final, resolving measure.

Second, even within the standard model, a theory that ignores gravity, thorny technical issues that are associated with quantum processes are swiftly solved if the theory is supersymmetric. The basic problem is that every distinct particle species makes its own contribution to the microscopic quantum-mechanical frenzy. Physicists have found that in the bath of this frenzy, certain processes involving particle interactions remain consistent only if numerical parameters in the standard model are fine-tuned—to better than one part in a million billion—to cancel out the most pernicious quantum effects. Such precision is on par with adjusting the launch angle of a bullet fired from an enormously powerful rifle, so that it hits a specified target on the moon with a margin of error no greater than the thickness of an amoeba. Although numerical adjustments of an analogous precision can be made within the standard model, many physicists are quite suspect of a theory that is so delicately constructed that it falls apart if a number on which it depends is changed in the fifteenth digit after the decimal point.5